

A precision viticulture UAV-based approach for early yield prediction in vineyard

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Abstract

Early yield prediction is a key factor for vineyard management optimization, to obtain the desired grape production and quality. Ground observation and manual weighing are time consuming and frequently provide low representative data. The aim of the study was to develop an automated early yield estimation system (5 weeks before harvest) using high-resolution RGB images, acquired through an unmanned aerial vehicle (UAV) platform in a representative zone of vigour variability of the whole vineyard. An unsupervised recognition algorithm was applied to derive the number of clusters and size, which have been used to estimate production. This fast and accurate methodology, which operated with a low cost setup, has shown high accuracy in yield prediction, providing interesting potential to support grape production management both in vineyard and in cellar.

Keywords: UAV, image analysis, grape detection, early yield prediction, low-cost

Introduction

Knowledge of spatial variability across a field plays a key role in vineyard management, affecting a wide range of aspects related to crop development (Bonilla *et al.*, 2015). Among those, grape yield is an important component that drives the return of investment in wine production (Van Leeuwen and Darriet, 2016) and its estimation is a challenging process. Moreover, yield prediction can improve the efficiency both for harvest operations in the field and winemaking management (Cunha *et al.*, 2010; De la Fuente *et al.*, 2015). Forecast prediction is a key aspect for farmers but yield variability across years, makes a correct estimation between different seasons difficult (Sabbatini *et al.*, 2012). Traditional yield estimation is realized by taking measurements manually in the field and extrapolating to the entire vineyard – making it extremely time consuming, destructive and subjective (Nuske *et al.*, 2014). Furthermore, there is a high risk that a limited number of samples fails to properly characterize spatial variability within the vineyard, thus compromising the data (Tisseyre *et al.*, 2018). In this context, the emerging technologies applied in agriculture provide new solutions to overcome the weaknesses of traditional ground measurements (Dhondt *et al.*, 2013; Di Gennaro *et al.*, 2017; White *et al.*, 2012). Viticulturists have developed different systems for yield estimating. De La Fuente *et al.* (2015) provided a performance review of six yield prediction methods applied to different phenological stages, identifying fruit set and veraison as best solutions in terms of earliness of prediction in the growing season, accuracy and precision.

In the last decade, advances in sensor technologies and image computing software provide a valuable tool to overcome those issues. As summarized by Aquino *et al.* (2018), different approaches have been tested to realize a partial or total unsupervised cluster for berry detection by high resolution RGB images. The most widely used approaches to analyse pixel colours is the RGB colour space based on three colour components which are red (R), green (G), and blue (B). However, several colour spaces have been introduced to exploit their mathematical properties (Khattab *et al.*, 2014). Among different colour spaces, the L*a*b* and HSV are the most frequently used (Bora *et al.*, 2015). Some authors have reported studies of image processing techniques taken manually in a vineyard (Liu and Whittey, 2015; Aquino *et al.*, 2017; Mack *et al.*, 2017). Other recent studies have reported

methodologies for large-scale yield monitoring in the field, by means of on-the-go image acquisition using modified terrestrial vehicles and 2D image analysis approaches (Nuske *et al.*, 2014; Font *et al.*, 2015; Aquino *et al.*, 2018). Those solutions have acquired images at night using artificial illumination on defoliated cluster zones, which provide accurate yield detection. However, they are time-consuming due to low forward speed especially in night condition. The aim of this study was to implement a fast and unsupervised methodology for yield prediction several weeks before harvest with reduced time and costs in terms of acquisition and processing of data. To obtain early automated yield estimation, high resolution RGB images were taken with a low cost UAV platform in a representative zone of vigour variability in a vineyard.

Materials and methods

Experimental site

The research was carried out in a 1.4 ha vineyard (43°25'45.30"N, 11°17'17.92"E) planted in 2008 and located on a slight southern slope at 355 m above sea level near Siena (Italy). Sangiovese cv. (*Vitis vinifera*) vines are trained to a vertical shoot-positioned trellis and spur-pruned on a single cordon with 2.2×0.75 m vine spacing and NW-SE rows.

Image acquisition

UAV technologies were chosen considering their potential in term of flight speed, low quote operative flight and custom angle of image acquisition. Remote aerial surveys were performed using an open-source UAV platform consisting in a modified multi-rotor MikrokopterOktoXL (HiSystems GmbH, Moomerland, Germany). The system was equipped with a multispectral camera ADC-Snap (Tetracam Inc., CA, USA) and a RGB camera Sony Cyber-shot DSC-QX100 (Sony Corporation, Tokyo, Japan). The UAV flight campaign was conducted in the experimental vineyards on 26 June and 07 August 2018. A preliminary survey on first date was performed equipping the multispectral camera to analyze vigour spatial variability, and to identify 2 representative zones, one in low vigour (LV) and one in high vigour (HV). A second multispectral survey was undertaken on 07 August to confirm that a vigour representative zone had been chosen, and produce the yield map. Multispectral image acquisition was at 50 m AGL, yielding a ground resolution of 0.025 m/pixel and 70% of overlap in both directions. During the second flight date, 8 consecutive vines within each representative zone were defoliated at fruit zone level. A RGB survey was carried out at 14.00 PM with 5 m AGL flight quote, with oblique camera (45°) along the row, providing a ground resolution of 0.002 m/pixel on the fruit zone. A series of RGB images of vines were acquired hovering about 20 s in front of each 8-vine plot. Sentinel-2 images were freely downloaded from the Copernicus Scientific Data Hub web site as a Level 2A product. This level is ortho-rectified and UTM geocoded, bottom-of-atmosphere multi-spectral reflectance.

Image processing

Multispectral images acquired by UAV were processed using Agisoft Photoscan Pro 1.1.6 (Agisoft LLC, St. Petersburg, Russia) to produce the ortho-mosaic and the digital elevation model (DEM). Matlab software (The MathWorks, Inc., Natick, USA) was employed for radiometric calibration based on reference panels and row filtering (Matese *et al.*, 2016). Next, NDVI data were extracted at single vine level with Quantum GIS (QGIS – <http://www.qgis.org/>), to correlate vigour and yield of each sample vine. NDVI from Sentinel 2 imagery was calculated using near-infrared B8 band (842 nm) and red B4 band (665 nm), producing a 10-m spatial resolution map.

The image processing workflow is shown in Figure 1. A first supervised step has been done to identify best image fitting on target vines for each vigour zone and to cut the sample vines region. Then, ImageJ software was used to perform an unsupervised processing of selected images. At this point, each original RGB image was converted to the three layers HSV color space: H

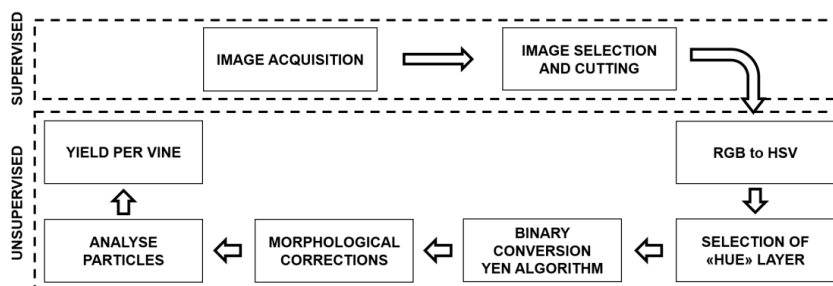


Figure 1. Image processing workflow.

(hue) distinguishes among the perceived colors, S (saturation) refers to how much of the light is concentrated at each specific wavelength of the hue; and V (value) represents the total brightness. The H layer was selected for image analysis, since it has the best performing layer for cluster segmentation. A threshold and two different filter methods were now applied to the H layer. (1) The Yen filter (Yen *et al.*, 1995) converted the layer to a 8 bit binary image, providing a black and white image with clusters segmentation from other objects (soil, leaf, trunk). (2) A median filter eliminated 'salt and pepper' noise, replacing each pixel with the median value in its 3×3 neighborhood. (3) A filling holes filter removed the small holes produced by the threshold based segmentation step. Finally, the command 'Analyze particles' was used to count and measure each cluster area detected in the binary 2D image.

Evaluation of yield prediction map

Yield estimation was carried out using the number and dimensions of detected clusters obtained from image analysis. First, an average yield per vine was calculated in each vigour zone from UAV images by converting total cluster area per vine (cm²/vine) from UAV images in yield per vine (kg/vine), on the basis of calibration parameters obtained from previous work in the same vineyard (2017 season). The yield per vine within different vigour zones were then correlated with NDVI data extracted with a polygon centered on each sample vine from 07 August 2018 NDVI map. The linear estimation model was applied to produce a yield prediction map based on vigour spatial variability in the vineyard. The same approach was tested with the Sentinel 2 NDVI map acquired closest to cluster detection survey (18 July 2018).

Results and discussion

Following the workflow described above, the processing of each image required a maximum of one minute. Figure 2 shows graphical flow output applied in LV and HV zones, beginning with supervised selection and cutting of image of sample vines (Figure 2A,B), and then RGB to HSV transform (Figure 2C), and clusters data extraction from binary image corrected with morphological filters (Figure 2D).

Data analysis showed a strong linear correlation between estimated yield and observed yield in terms of ripe cluster per vine ($R^2=0.84$) (Figure 3A). The evaluation of the protocol response against ground traditional measurements showed a strong linear relationship during early survey at the beginning of August in mid veraison stage. However, unripe and partial overlapping conditions affect the absolute value detected, leading to an error of 17.8% in LV and 23.3% in HV in respect to ripe yield per vine, and higher concerning the total yield per vine (LV=30.5% and HV=27.8. Seasonally variability strongly affects cluster features, so high differences in terms of cluster morphology, berry ripeness within the cluster, cluster overlap level, canopy development and light condition could occur

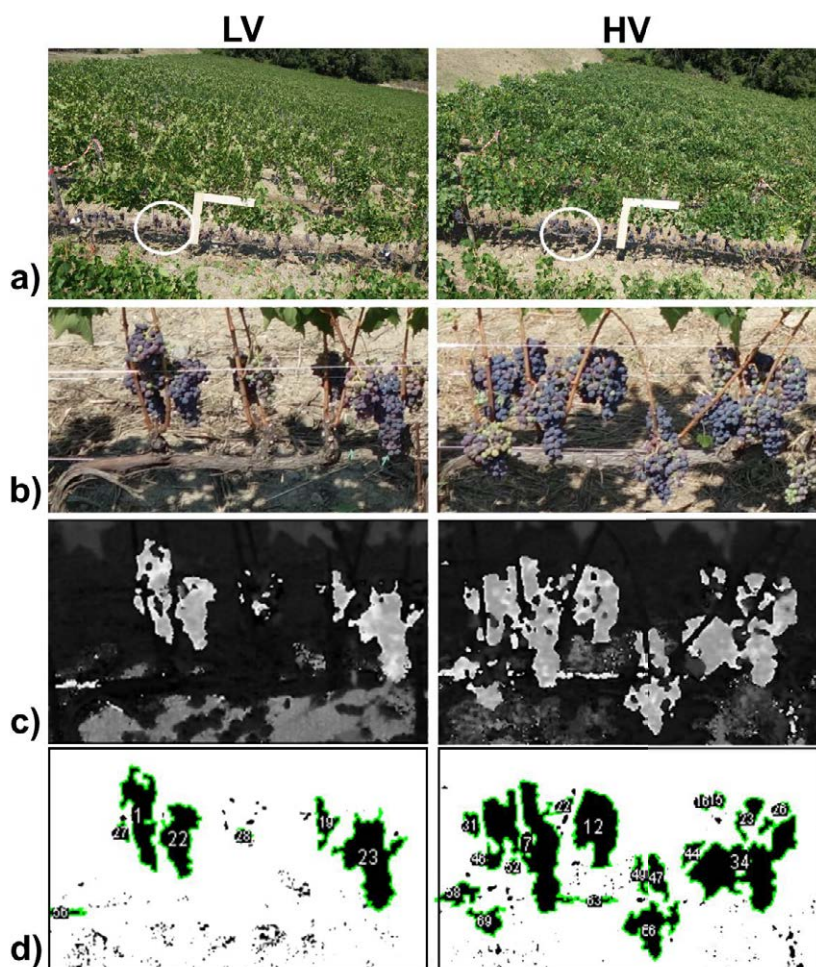


Figure 2. Example of image analysis chain from supervised (A, B) to unsupervised (C, D) step on low (LV) and high (HV) vigour sample vines: (A) original image, (B) sample vine detail; (C) hue layer of HSV color space; (D) binary image based on Yen threshold algorithm with morphological filter and clusters area extraction.

between different seasons. Consequently, it would be interesting to investigate the methodology performance across different seasons, and, finally potentially identify a seasonal coefficient to correct yield estimation.

However, errors in yield prediction are highlighted in other on-the-go research at harvest time with fully ripe grape: Font *et al.* (2015) achieved 16% prediction error while Nuske *et al.* (2014) reported yield prediction errors between 6.5-11.7%. Taking into account an early approach, Aquino *et al.* (2018) described the highest performance methodology reaching a 12.83% of error before veraison. As reported in other work on yield monitoring systems in vineyard (Taylor *et al.*, 2016), the methodology was confirmed to be effective for characterization of relative spatial yield patterns but, to improve absolute yield data, a seasonal adjustment is necessary against ground measurement on sample vines before it can be applied in large scale yield estimation. In regards to absolute mean

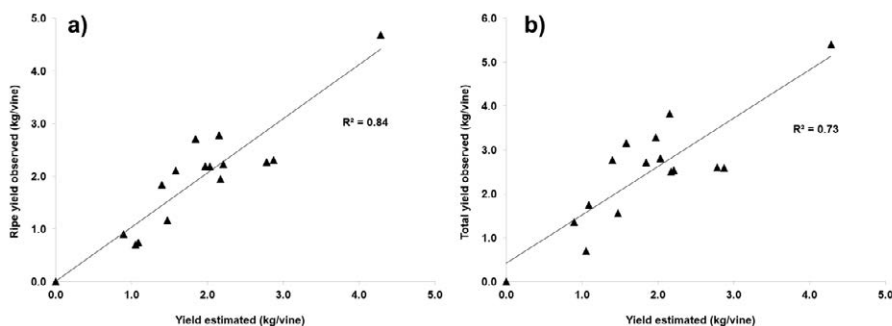


Figure 3. Estimated yield correlated with observed ripe cluster yield per vine (A), and observed total yield per vine (B).

values in different vigour zones, a discrete presence of partially overlapping between clusters in HV vines caused a mean yield under-estimation of the methodology (HV=2.33±0.97 kg, LV=1.78±0.69 kg) with respect to ground observation (HV=2.57±1.07 kg, LV=1.74±0.64 kg). Observing the performance in total yield estimation monitored at ground (HV=3.00±1.43 kg, LV=2.14±1.06 kg), the methodology loses all clusters that are partially or fully unripe, providing interesting linear correlation ($R^2=0.73$) but a higher under-estimation (Figure 3B). Undertaking a survey close to harvest, could reasonably achieve higher performance also in final yield estimation. Furthermore, an increase in berry weight from the beginning of August (UAV survey) could be another cause of the slight yield under-estimation compared to that measured at the harvest time 5 weeks later. Subsequently, NDVI data were extracted for sample vines in both vigour zones and correlated with the ripe yield per vine estimated by UAV approach. The linear regression between estimated yield of ripe grapes and NDVI extracted for each vine allowed a 'maturity yield map' generation based on NDVI spatial variability within the vineyard. A two-class yield map was generated based on the median distribution of NDVI values between high and low vigour (HV=2.84±1.20 kg/vine, LV=2.05±1.05 kg/vine) (Figure 4A). The same approach has been tested on a Sentinel 2 NDVI map

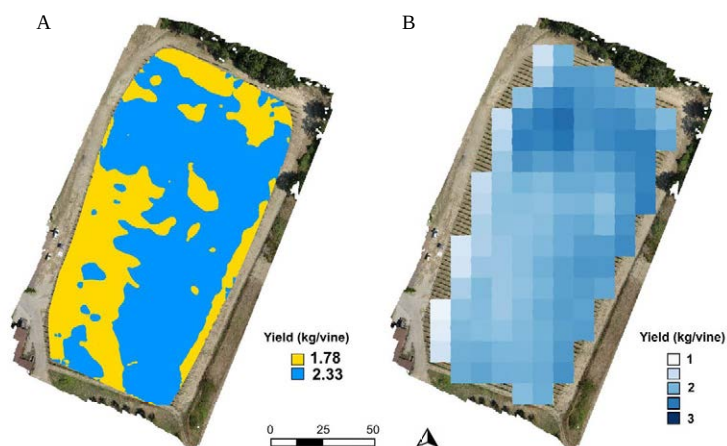


Figure 4. Maturity yield map based on vigour spatial distribution in UAV (A) and Sentinel 2 (B) NDVI map.

through correlation from NDVI pixel values containing 8 sample vines and mean estimated yield per vine providing similar results (Figure 4B). Those maps represent a snapshot of the ripening process within the vineyard to the farmer at the time of flight. Based on the knowledge of the vineyard potential yield, this tool could provide fast information for an accurate yield forecasting or eventually modify the cluster load with thinning practices to achieve yield and quality desired. Considering the short time needed to move within the vineyard, acquire images of fruiting zones and process images, the increase of the sampling size will add minimal impact in respect to a high improvement in accuracy. In fact, taking into consideration the possibility of monitoring also intermediate vigour zone, the methodology would be able to realize yield maps with greater prediction performance. However, given that the study vineyard is characterized by a clear distinction of vigour, the separation into 2 classes was sufficient.

Low-altitude inclined photogrammetry allows the acquisition of extreme detailed images of the fruiting zone. A primary requirement on the application of this method concerns the line of sight between sensor and cluster that has to be free to assure a good acquisition. However, the common practice of partial leaf removal a few weeks before harvest, to optimize ripening and preserve grape health, is a way to overcome this requirement. Furthermore, the automatic procedure based on color thresholds cannot discriminate green clusters from leaves, and therefore only partially and totally ripe clusters are identified. Except for these aspects, there are many advantages of the methodology. Firstly, the speed of acquisition and processing, which exploits single images taken in representative areas that are quickly reached by the UAV. This is a highly transferable approach, as it uses low-cost hardware and open source analysis software. Furthermore, the remote sampling approach based on spatial variability acquired from farmer knowledge and/or vigour map (high resolution UAV or free low resolution satellite) ensures representative data especially in large areas or highly fragmented and sloping vineyards. Although other on-the-go methods perform better in terms of accuracy in yield estimation, another strong point is the timeliness of acquiring images intra- and inter-fields, data processing and output report as simple spatial information or yield map. The proposed method also has high flexibility of application providing an automated way to acquire information such as leaf area or chromatic alterations caused by leaf symptomatology (biotic and abiotic stress). The methodology is not intended to replace visual scouting on the ground, but as a tool to support scouting with potential for timely monitoring of large areas.

Conclusions

The innovative approach shows an interesting potential in practical application both for farmer support in grape production management both in vineyard and in cellar, and for service companies. The development of a yield estimation protocol based on low cost hardware (UAV and RGB camera) and software (open source image analysis tool), with high level of automation, provides a flexible and user-friendly solution for yield monitoring. The methodology developed has considerable potential in terms of recognizing the number of clusters and quickly estimating ripe yield at the time of flight; however, it suffers from an unavoidable physical limit represented by leaf cover.

The proposed approach requires further investigation to evaluate performance in different seasons, varieties and training systems, although an obvious limit remains the need to have a free line of sight between sensor and clusters.

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